

Yassine Machta

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Education

- **ENSAE Paris** 2021–2025
Engineering Curriculum: Data Science, Statistics and Learning. GPA: 3.8
- **Institut Polytechnique de Paris** 2024–2025
Master 2 Data Science
- **Lycée Pierre de Fermat** 2019–2021
*Prepa MPSI/MP**

Work Experience

- **Research Engineer** Inria - [ARTICULAB](#), Paris
Jan 2026 – Jun 2026
 - **Real-time gesture generation:** Building a data-driven model to produce co-speech gestures for the team's embodied conversational agent.
- **Research Intern** Inria - [ARTICULAB](#), Paris
May 2025 – Nov 2025 *Supervised by: [Cassell, J](#) and [Le Chapelier, M](#)*
 - **Gesture retrieval engine (NLP + embeddings):** Built a real-time system for an embodied conversational agent combining rule-based heuristics and embedding similarity to select contextually appropriate gestures. Published at ACM MM GENEVA 2025.
- **Research Intern** Sony R&D, Stuttgart
Jan 2024 – Jun 2024 *Supervised by: [Facijs, Z](#) and [Franceschi, R](#)* *Letter of Recommendation*
 - **3D reconstruction pipeline:** Developed an automated end-to-end pipeline for image preprocessing and 3D mesh reconstruction; evaluated the impact of uncontrolled capture conditions on output quality.
 - **Internal web app:** Built a tool to streamline mesh alignment and quality evaluation for engineering teams.
- **Research Intern** Inria - [SIMBIOTX](#), Saclay
May 2023 – Nov 2023 *Supervised by: [Ali, O](#) and [Vignon-Clementel, I.](#)*
 - **3D liver vessel segmentation (deep learning):** Implemented topology-preserving losses to improve segmentation accuracy; analyzed vessel tree morphometry. Published at MICCAI ADAMI 2024 ([Code](#)).
- **Data Science Intern** Cour des Comptes, Paris
Jun 2022 – Sep 2022 *Supervised by: [Grignon, P](#)*
 - **Network analysis (R):** Built descriptive statistics and cartography pipeline to map the French Airport Network for a national audit.

Skills

- **Programming Languages:** Python (PyTorch, Scikit-learn/image, Selenium, pandas, Numpy, matplotlib), Docker, Kubernetes, ArgoCD, Slurm, Git, GitHub Actions, CI/CD. learning CUDA/C++ and Godot.

Projects & Labs

- **LLM fine-tuning (LoRA + DPO):** Fine-tuned LLMs; reimplemented *Unsupervised Multimodal Translation* to learn transformer and multimodal architectures.
- **RL in game environments:** Trained agents to navigate environments and play Skull King using reinforcement learning.
- **Generative models (diffusion + flow matching):** Conditional graph generation; music sheet generation via diffusion and [Flow Matching](#).
- **GPU programming (CUDA):** Monte Carlo simulation for Variance Gamma model; rewrote and optimized PyTorch GPU kernels to reduce inference time for a torch model.

Publications

- **Machta, Y.**, Le Chapellier, M., Silem, O., Cassel, J (2025). "Semantic Gesture Retrieval for real-time co-speech gestures." *ACM Multimedia GENEVA Workshop*, 2025. ([Poster](#))
- **Machta, Y.**, Ali, O., Facque, A., Vlaschenau A., Hakkakian, K., Golse N., Vignon-Clementel, I. (2024). "Improving 3D Liver Vessel Segmentation through Topology Preservation." *MICCAI ADAMI Workshop*, 2024. ([Paper](#), [Poster](#), [Code](#))

Extracurricular Activities

- **ENSAE English Debating Team** 2022–2023
FDA Tournament Winner
- **Sports and Student Councils** 2022–2023
Organized various events and activities at ENSAE Paris. Played on the volleyball team.

Languages

- **Arabic:** Native
- **French:** Native
- **English:** C1

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